Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 27)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

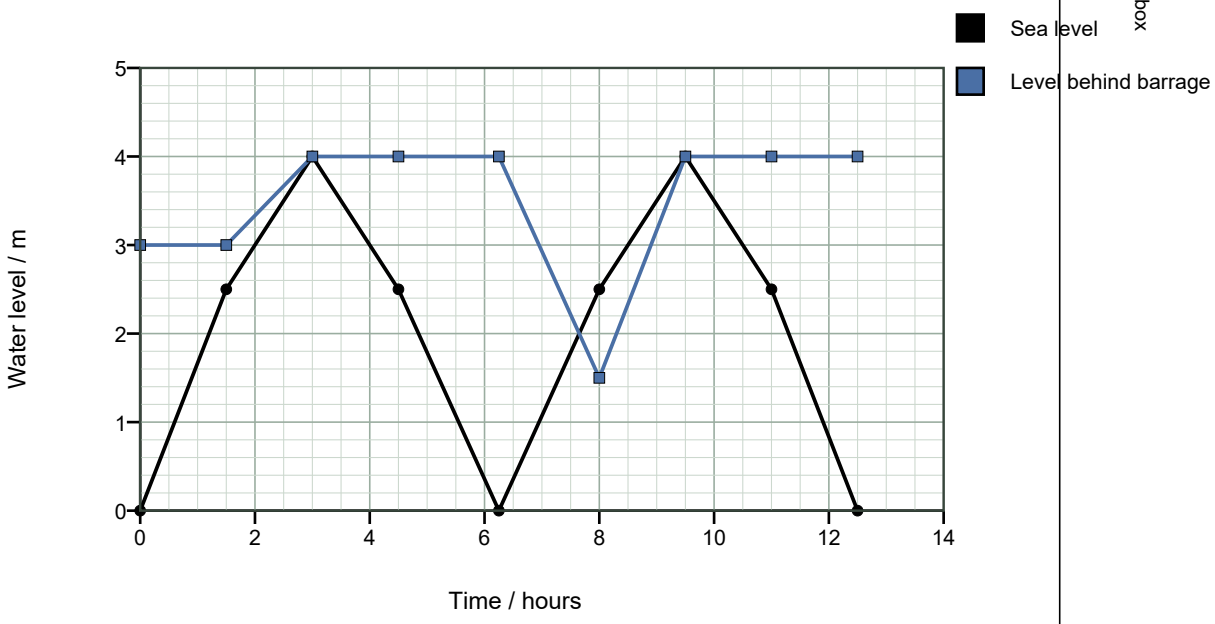
Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 1 . 1 Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

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0 1 . 2 Explain why tidal power is intermittent but predictable.

[2 marks]

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0 1 . 3 Explain the difference between spring and neap tides and why this affects output.

[2 marks]

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0 1 . 4 Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

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0 1 . 5 Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

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0 2 . 1 Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

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0 2 . 2 Explain how noise can damage non-living objects.

[3 marks]

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0 2 . 3 Explain how noise can reduce the breeding success of birds.

[2 marks]

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0 2 . 4 Explain why military sonar is a threat to cetaceans.

[2 marks]

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0 3 . 1 Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation .....

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Urban development .....

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Agriculture .....

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0 3 . 2 Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

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0 3 . 3 Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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0 4 . 1 Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

0 4 . 2 Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

0 4 . 3 Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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0 5 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

0 5 . 1 Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

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0 5 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

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0 5 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

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0 5 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

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0 6 . 1 Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

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0 6 . 2 Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 6 . 3 Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

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0 6 . 4 Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

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0 7 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 7 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 7 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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0 8 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 8 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 8 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 9 Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

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0 9 . 1 Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]

Answer ☐ %

0 9 . 2 Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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0 9 . 3 Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

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0 9 . 4 Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

11 . 1 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

11 . 2 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS